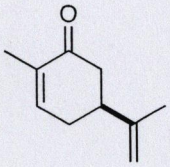
Organic Cumulative Exam Sep 8, 2017

Topic: Physical Organic Chemistry

All of the questions below can be answered with a few words or structure(s). Please be concise in your answers.

1. Identify 3 things that are missing from the representation of (S)-carvone shown that actual carvone molecules possess.



2. What information does the solution to the Schödinger equation provide about a molecule, e.g. (S)-carvone?

3. What does an atomic orbital represent?

4. Draw 3 examples of sigma bonds and 2 examples of pi bonds using s, p, or d atomic orbitals.

5. Draw the lowest energy structure of ethyl cation.

6. Draw the structure of the product of the gas phase reaction of methane and a proton: $CH_4 + H^+ \rightarrow$?